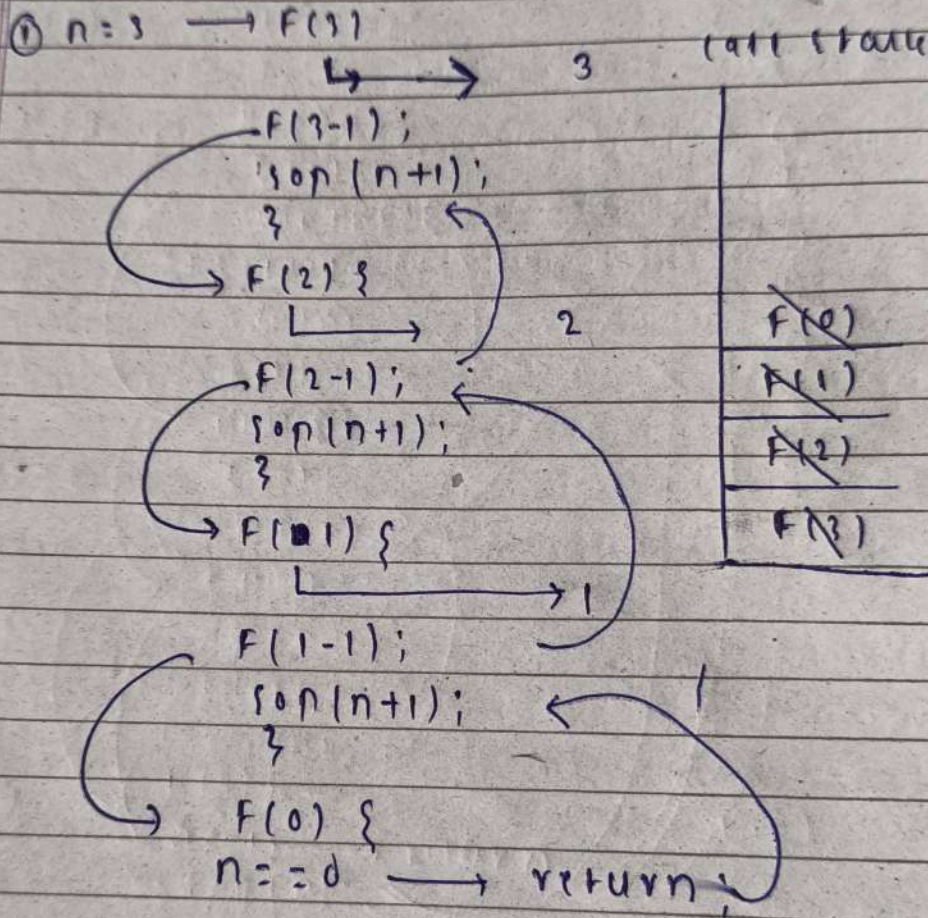


```

① void fun (int n) {
    if (n == 0) return;
    cout << n << " ";
    fun(n-1);
    cout << (n+1) << " ";
}
fun(3);
    
```



Final output: -

3	2	F(2)
2	3	/
1	4	F(3)
2		
3		F(1)
4		
		F(0)

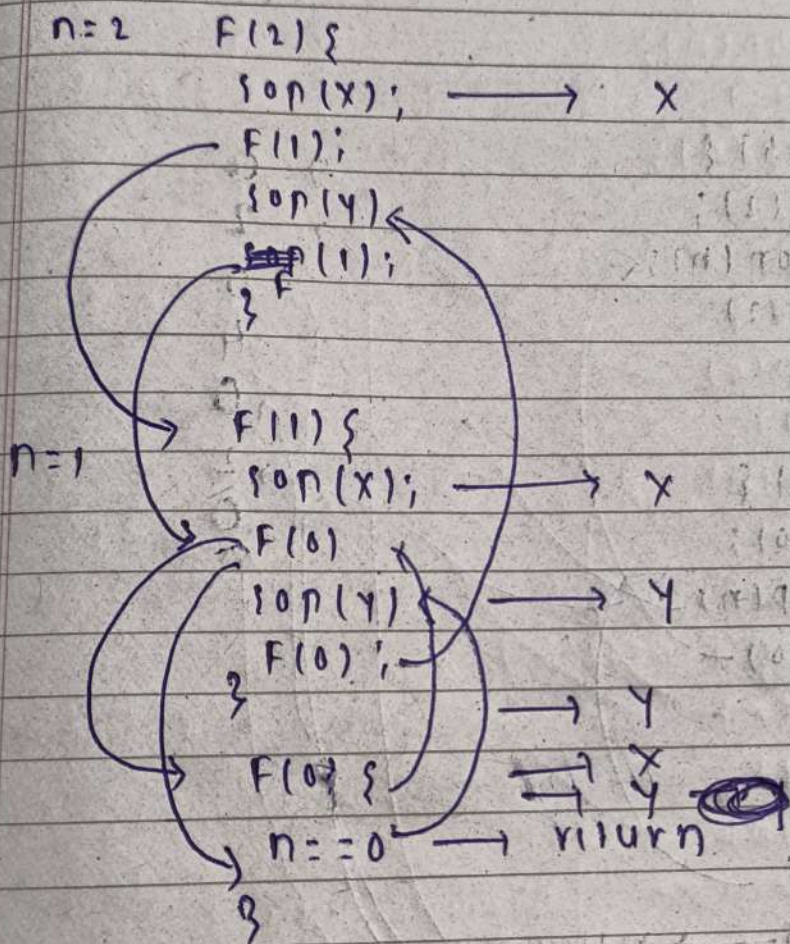
2)

```
void fun(int n) {
    if (n == 0) return;
    cout << "X" << endl;
    fun(n-1);
    cout << "Y" << endl;
    fun(n+1);
}
```

→ call fun(2)

call stack

F(0)
F(0)
F(1)
F(0)
F(0)
F(1)
F(2)



→ Final output →

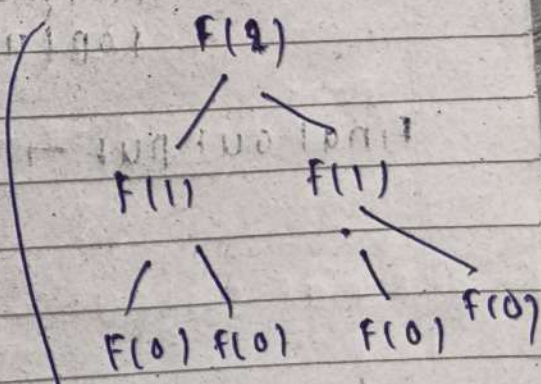
X

Y

Y

X

Y

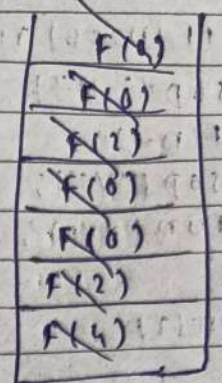



```

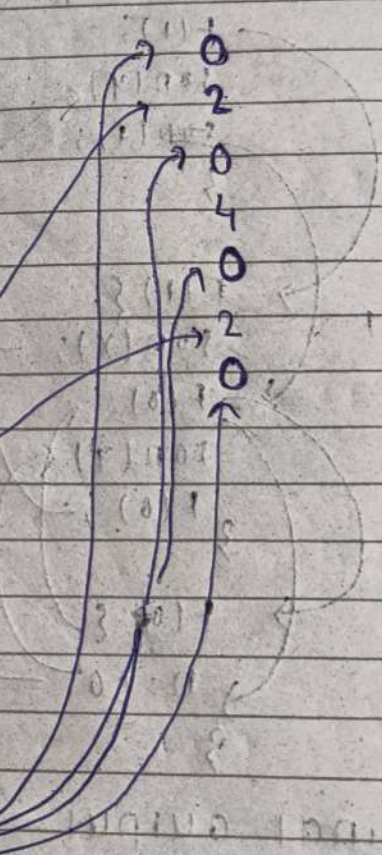
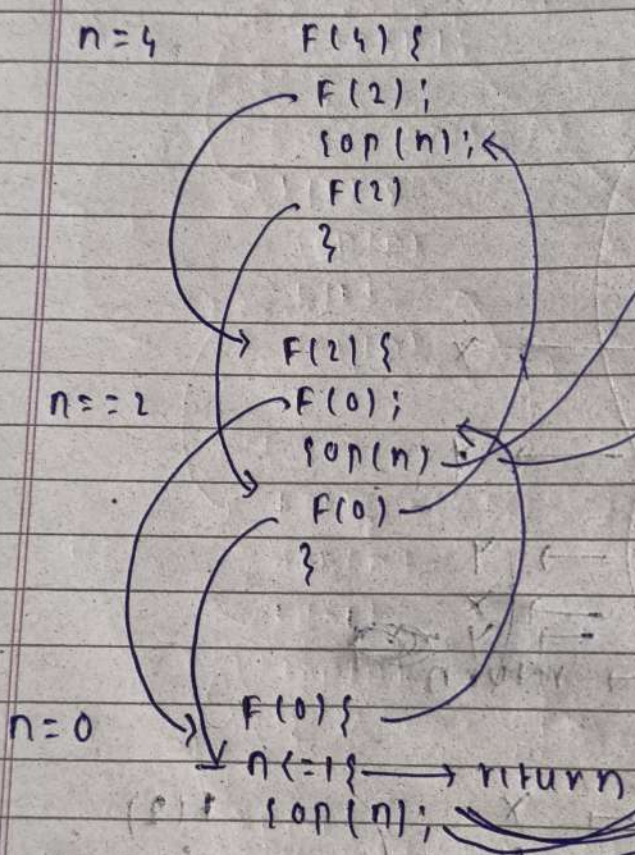
(3) void fun(int n) {
    if (n <= 1) {
        top[n];
        return;
    }
    fun(n-2);
    top[n];
    fun(n-2);
}

```

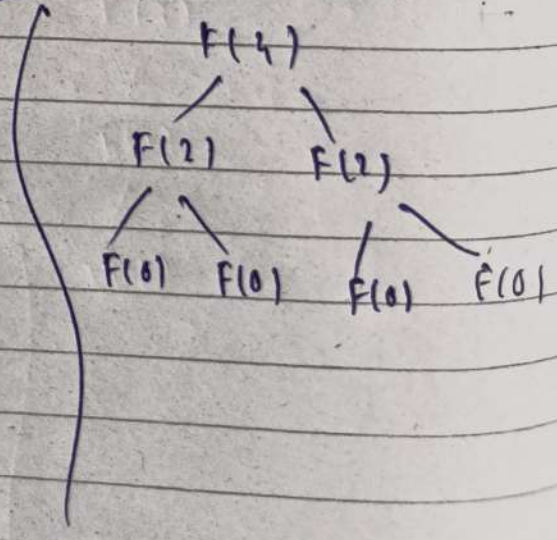
call stack



call
→ F(4);



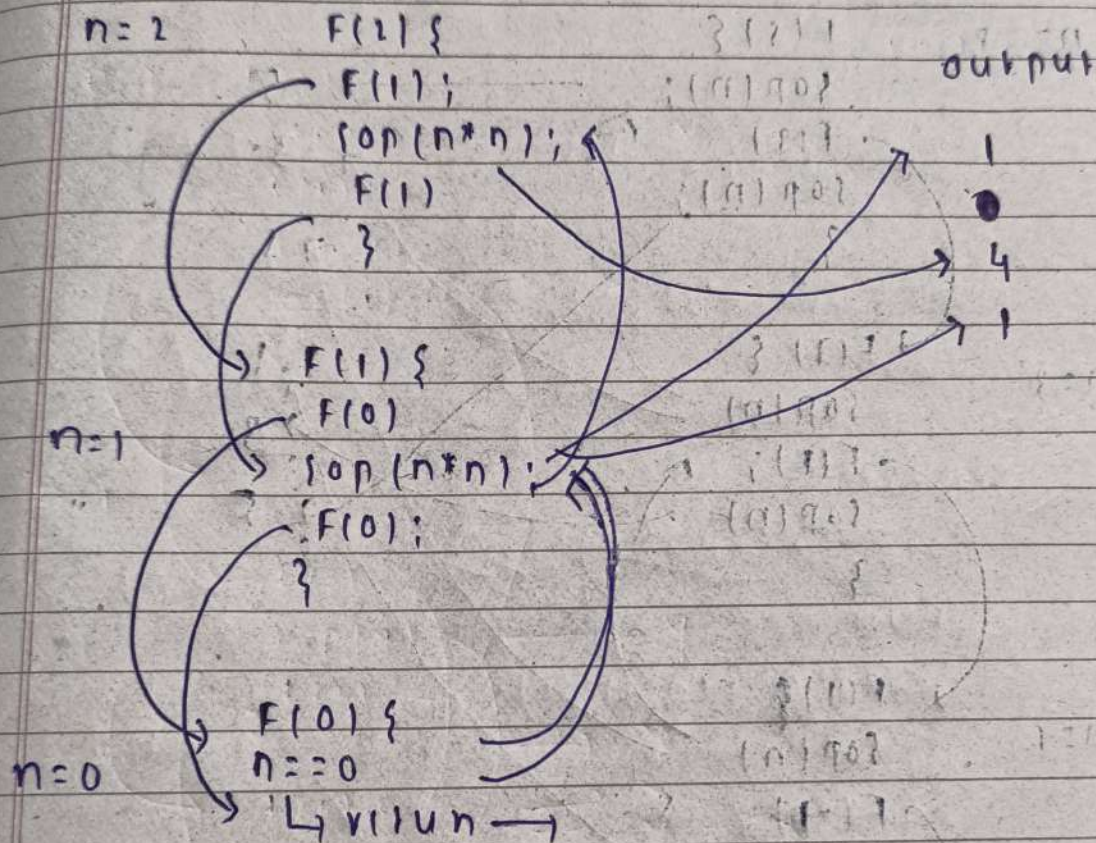
Final output →
 0
 2
 0
 4
 0
 2
 0



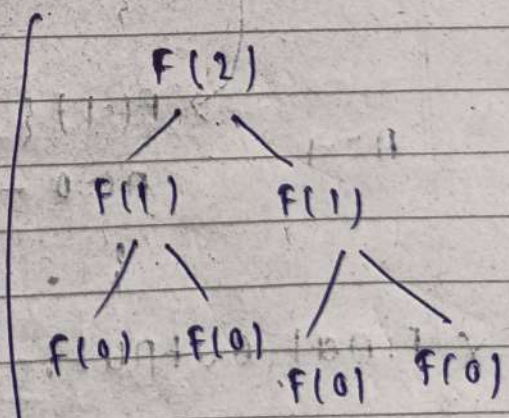
④ void fun(int n) {
 if (n == 0) return;
 fun(n-1);
 cout << n * n << " ";
 fun(n-1);
}

call → F(2)

call stack
~~F(0)~~
~~F(0)~~
~~F(1)~~
~~F(0)~~
~~F(0)~~
~~F(1)~~
~~F(2)~~



Final output → 1
4
1



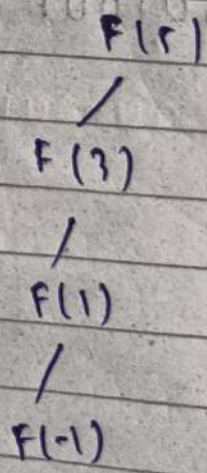
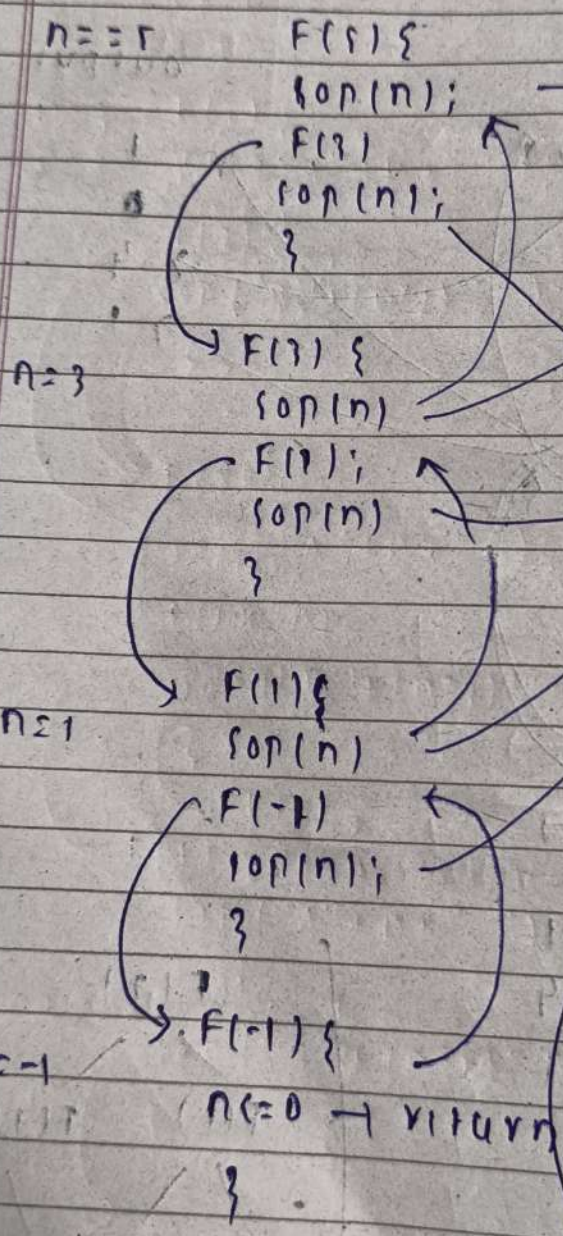
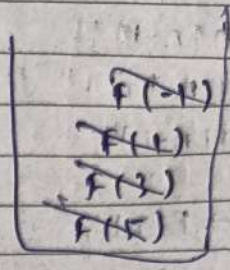
Call Stack

```

void fun(int n) {
    if (n == 0) return;
    {
        sop(n);
        fun(n-1);
        sop(n);
    }
}

```

→ call fun(5)



x Final output:-

c) void fun(int n) {
 if (n == 1) {
 top(n);
 return n;
 }
 top(n);
 fun(n/2);
 top(n);
}

Call Stack

F(8)
F(4)
F(2)
F(1)

→ call → Fun(8)

n = 8

F(8) {
 top(n);
 F(4);
 top(n);
}

n = 4

F(4) {
 top(n);
 F(2);
 top(n);
}

n = 2

F(2) {
 top(n);
 F(1);
 top(n);
}

n = 1

F(1) {

n <= 1 → top(n);
 return n

Final output: -

8
 4
 2
 1
 2
 4
 8


```

(7) void Fun (int n) {
    if (n <= 0) {
        return;
    }
    cout << "A" << n << " ";
    Fun (n-1);
    cout << "B" << (n*2) << " ";
}

```

→ call → F(3)

```

n=3 F(3) {
    cout << "A" << n << " ";
    F(2);
    cout << "B" << (n*2) << " ";
}

```

```

F(2) {
    cout << "A" << n << " ";
    F(1);
    cout << "B" << (n*2) << " ";
}

```

```

F(1) {
    cout << "A" << n << " ";
    F(0);
    cout << "B" << (n*2) << " ";
}

```

```

F(0) {
    n <= 0 → return
}

```

F(0)
F(1)
F(2)
F(3)

A 3

A 2

A 1

B 2

B 3

B 4

Final output = A 3

A 2

A 1

B 2

B 4

B 6


```

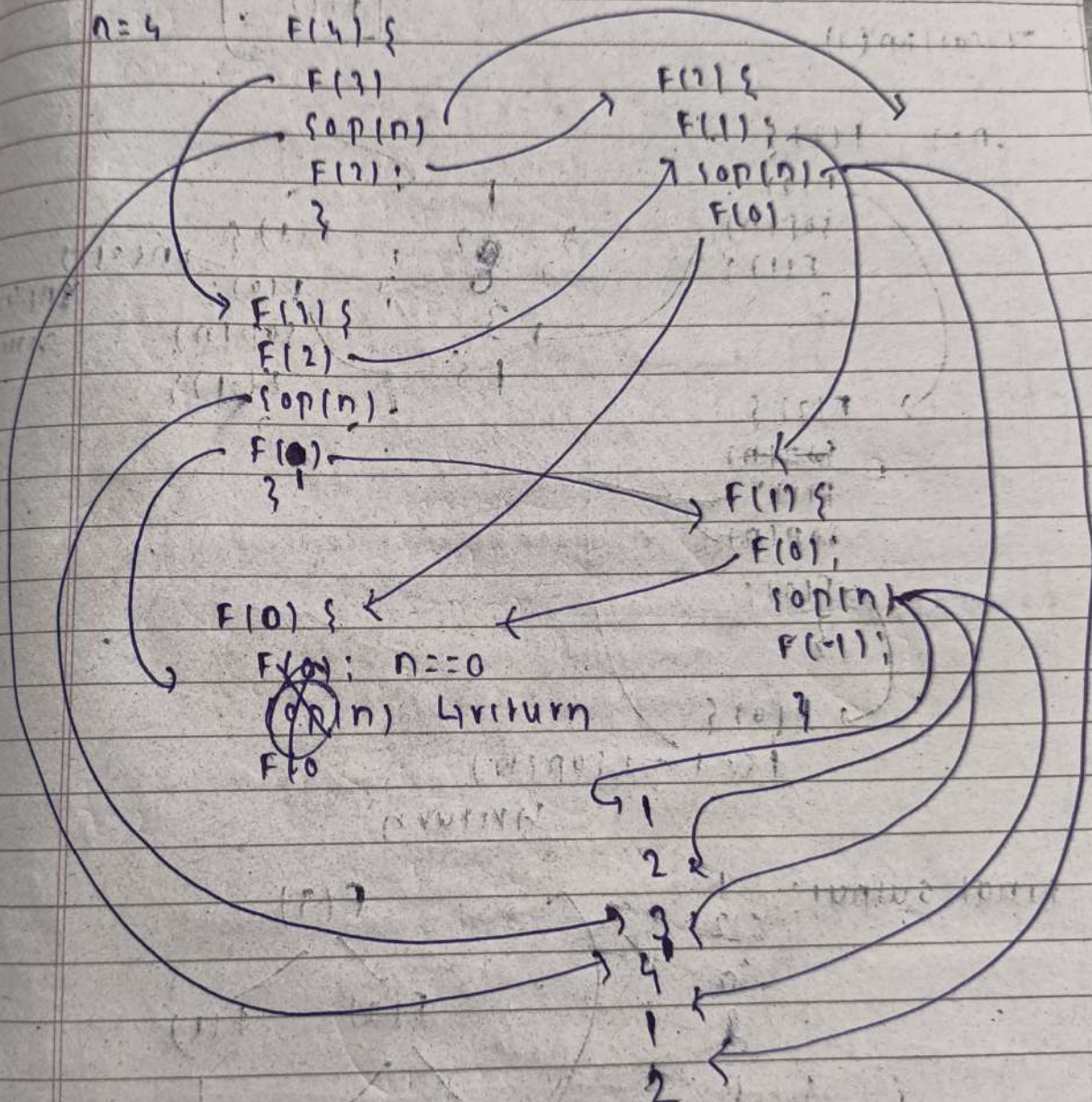
1 void Fun(int n) {
    if (n == 0) return;
    Fun(n-1);
    sop(n + " ");
    Fun(n-2);
}
→ call Fun(4);

```

Date: / /

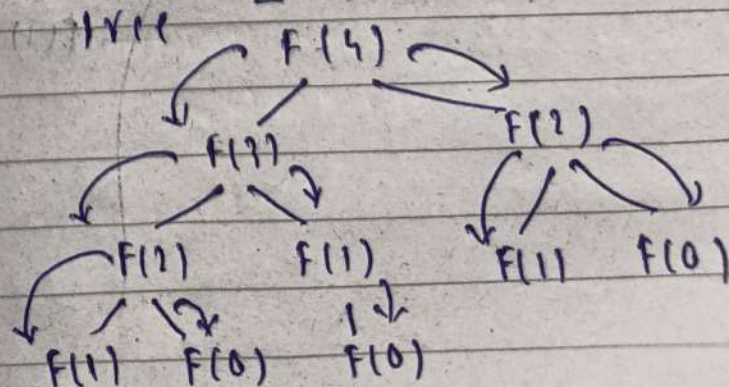
F(0)
 F(1)
 F(2) F(0)
 F(0)
 F(0)
 F(1)
 F(2)
 F(3)
 F(4)

n = 4



* Final →

1
 2
 3
 1
 4
 1
 2

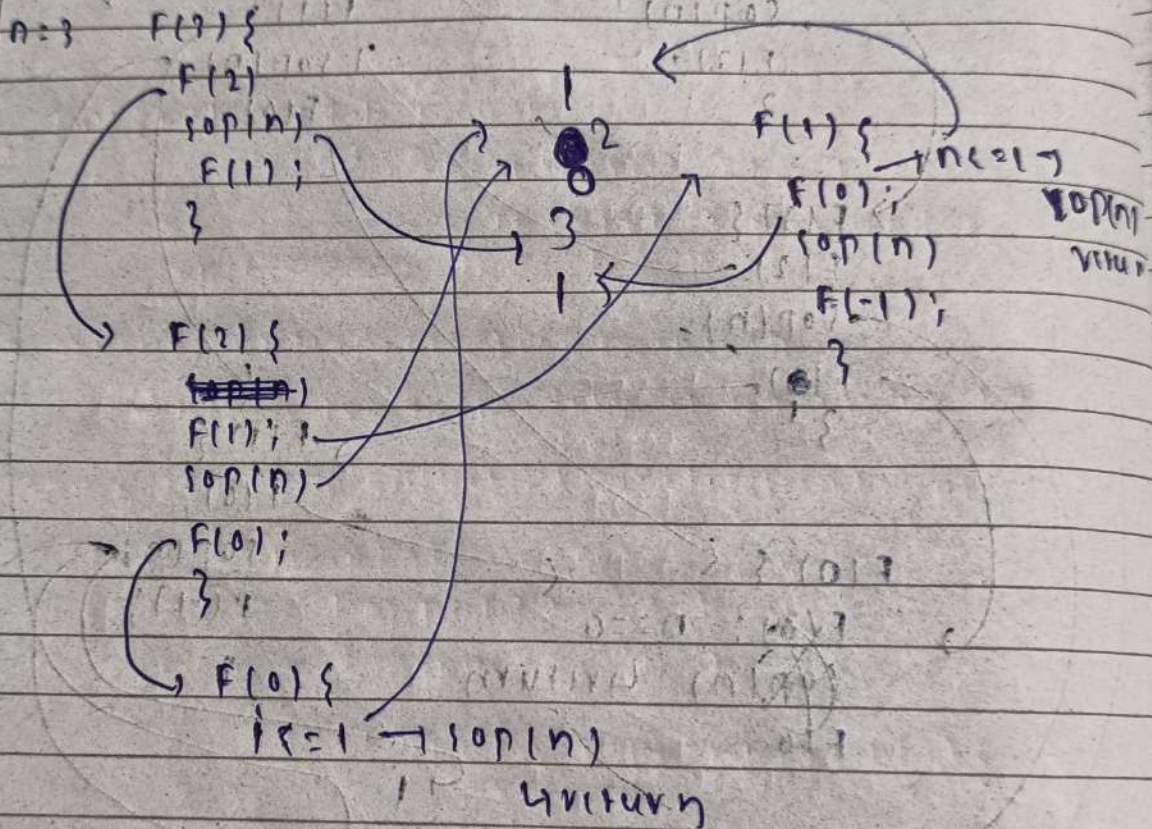



```

9) void fun(int n) {
    if (n <= 1) {
        sop(n + " ");
        return;
    }
    fun(n-1);
    sop(n);
    fun(n-2);
}

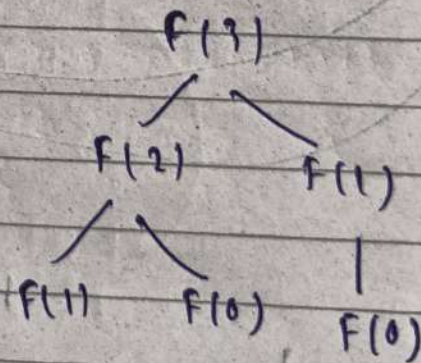
```

→ call fun(3)



Final output:

1
2
0
3
1



10

```
void Fun(int n) {
```

```
    if (n <= 0) return;
```

```
    Fun(n-1);
```

```
    F(n-2);
```

```
}
```

call $\rightarrow F(4)$

$n=4$

$F(4)$ {

$F(3)$

$F(2)$

}

$F(3)$ {

$F(2)$;

$F(1)$;

}

$F(2)$ {

$F(1)$

$F(0)$

$F(-1)$

}

$F(0)$

$F(-1)$

$F(0)$

call stack

$F(4)$

$F(3)$

$F(2)$

$F(1)$

$F(0)$

$F(-1)$

$F(0)$

$F(1)$

$F(2)$

$F(3)$

$F(4)$

$F(2)$ {

$F(1)$

$F(0)$

}

$F(0)$ {

$n=0 \rightarrow$ return

}

$F(-1)$ {

$n < 0 \rightarrow$ return

}

$F(4)$

$F(3)$

$F(2)$

$F(2)$

$F(1)$

$F(1)$

$F(0)$

$F(1)$

$F(0)$

$F(0)$

$F(-1)$

$F(0)$

$F(-1)$

$F(0)$

$F(-1)$